

ECE501 - Digital Image Processing

Weekly Report: Demorphing of Hybrid Images

Week 6

Group 10

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1. Overview of the Past Week

Week 5 covered the subject of color hybrid images. The Week 5 was devoted to the discovery and research of the perceptual illusions in full-colour images. We were concerned in discovering how the content of the colour channels at low and high frequencies has an influence on the perception of human beings and how illusion can be enhanced with sufficient colour-space processing.

2. Work Completed During the Week

- We reviewed the sources that dealt with demorphing and image decomposition.
- Looked deeper into frequency-based and gradient-based algorithm to separate high- and low-frequency layer in a picture.
- And performed early experiments of demorphing using Fourier Transform and Difference of Gaussian (DoG) filter.

3. Key Observations

- Demorphing is very much dependent on the frequency difference content in the hybrid image.
- It is possible to extract high-frequency details in part by using the edge enhancement and DoG filtering operations.
- Separation quality improves when hybrid images are created through control parameter of the Gaussian filter.

5. Work Planned for Next Week

- Try and test with machine learning-based demorphing with high-quality demorphing to compare the results.
- Test frequency masking and multi-level wavelet decomposition such that more precise separation may be done.

6. Conclusion

The time of this week was spent on the exploration of the ways on how to de-morph with the illusion of the hybrid picture and attempted to reverse it. The frequency-based filtering separation experiment enabled us to get a better insight into the limitation and challenges of separating images.